

Vilmodrine Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-836/C

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Cleaning verification swab results were below the health-based exposure limits derived for the product. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Deviations observed during the campaign were investigated and closed with no impact to product quality. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. The control strategy links critical quality attributes to critical process parameters identified during development. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

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Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

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Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The control strategy links critical quality attributes to critical process parameters identified during development. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

3.2.P.4 Excipient Control Strategy

For the commercial formulation F-836/C, Ligamed MF-2-V (Magnesium stearate, lubricant) is used.

Formulation Development Rationale

The formulation development rationale for Ligamed MF-2-V rests on robust tablet hardness across the compression force design space.

The approach described herein is consistent with FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms.

Registered Specification

Component	Function	Grade per current specification
Magnesium stearate	lubricant	Kosher Passover grade HyQual
Reference		SPEC-EX-1246 Rev 03

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

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No changes to the approved analytical procedures are proposed in this submission. The control strategy links critical quality attributes to critical process parameters identified during development. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

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Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable

food-contact regulations and compendial requirements for plastic packaging systems.

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